

Exercise Sheet for Week 10

Key:

- **Questions marked with a 🍷** are for discussion in your tutorials. Please complete these questions before your tutorial.
- **Questions marked with a 🦋** are difficult questions.
- **Questions marked with a 🎵** have been marked so that they don't feel left out. Please don't leave them out during your independent learning and revision time.

🍷 Exercise 1.

- (i) Show that $\ln\left(1 - \frac{1}{k^2}\right) = \ln\left(\frac{k+1}{k}\right) - \ln\left(\frac{k}{k-1}\right)$.
- (ii) Show that $\sum_{k=2}^{\infty} \ln\left(1 - \frac{1}{k^2}\right)$ is a telescoping series.
- (iii) Hence state $\lim_{n \rightarrow \infty} S_n$ where $S_n = \sum_{k=2}^n \ln\left(1 - \frac{1}{k^2}\right)$ is the n th partial sum.

🍷 Exercise 2. Determine the radius of convergence R of the following series, and determine whether or not the series converges at the end points $\pm R$.

(i) $\sum_{k=1}^{\infty} \frac{x^{2k}}{3^k}$ (ii) $\sum_{k=1}^{\infty} kx^k$ (iii) $\sum_{k=1}^{\infty} (5x)^k$ (iv) $\sum_{k=1}^{\infty} \frac{x^k}{1+k^2}$

🍷 Exercise 3. Find the Maclaurin series of the following functions. State the range of values of x for which these expansions are valid, and hence the radius of convergence for each series

(i) $f(x) = e^{-2x}$ (ii) $f(x) = (1+x^2)^{-1/2}$ (iii) $f(x) = \ln(1+x^2)$

🎵 Exercise 4. Discuss whether the following series converge. Do any of these series converge absolutely?

(i) $\sum_{n=1}^{\infty} \frac{e^{-n}}{n^2}$ (ii) $\sum_{n=1}^{\infty} \frac{3^n n^2}{n!}$ (iii) $\sum_{n=1}^{\infty} \frac{(-1)^n \ln n}{\sqrt{n}+1}$ (iv) $\sum_{n=1}^{\infty} \frac{4^{2n-1}}{3^{2n+1}}$

🎵 Exercise 5. Find the first three non-zero terms in the Maclaurin series for the following functions

(i) $f(x) = \frac{4}{3+e^x}$ (ii) $f(x) = \tan(x)$

🦋 Exercise 6. Show that $\arcsin(x)$ has the infinite Maclaurin series expansion

$$\arcsin(x) = \sum_{p=0}^{\infty} \frac{4^{-p}(2p)!x^{2p+1}}{(2p+1)(p!)^2}.$$